SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Modern Day Robotics & Its Industrial Applications (MDRIIA - 702CO0E012) · B.Tech Computer Science & Business Systems (CSBS), in partnership with TCS · AY 2026–27 Odd Semester

GROUP 10 TECHNICAL DOSSIER

Capstone ICA (20 marks): CDIO Robotics Engineering & MuJoCo Physics 10 · CSBS Techno-Economic Business Systems Analysis 5 · Studio Showcase & Viva Voce 5

Group · project	Group 10 · 7-DOF Surgical Robotic Assistant with Physiological Tremor Filtering for Precision Tasks
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026/tree/main/Group_10_Soumya_Harshal
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_10_Soumya_Harshal/docs/MDRIIA_Group_10_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
No slides	There is no PPT for MDRIIA. At the Expo and in your video you present the live MuJoCo simulation, your business sensitivity graphs, and a webcam picture-in-picture of every member.
How to fill this file	Use Microsoft Word (desktop). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them and do not edit this file in Google Docs: the fields are how your answers are read.

Student name	Roll No.	SAP ID	Capstone role(s)	Specialisation (finalised titles sheet / monorepo)
Soumya Singh	E071	70362400085	Lead Robotics Systems Architect + Autonomous Navigation / Vision Specialist	Lead Surgical Kinematics, Damped Least Squares IK & MuJoCo Modeler
Harshal Khandekar	E033	70362400064	Telemetry QA Lead + CSBS Business & Techno-economic Systems Analyst	Digital Signal Processing, Tremor Modeling & Kalman Filtering Lead

Roll numbers and SAP IDs are from the official student list. Tell Dr. Nanade by Tuesday, 20-Oct-2026 if any name, ID or role is wrong. Groups with fewer than four members combine roles as shown.

SECTION 1 · Problem Bottleneck, Industry Standards & Aalborg Research Title *Component 1 · CDIO Robotics (10)*

Authorised Aalborg title	How can a 7-DOF surgical manipulator simulated in MuJoCo implement inverse kinematics Jacobian damping and low-pass tremor filtering to achieve sub-0.5 mm needle placement accuracy under simulated physiological surgeon hand tremor?
Original draft topic	Intelligent robotics assistant for precision surgical task
Why the title changed	The student submission was already of outstanding quality, explicitly naming 7-DOF kinematic redundancy, MuJoCo simulation, sub-0.5 mm precision, and stereotactic neurosurgical tremor mitigation. To elevate it to premier conference publication standards (e.g. IEEE BioRob or IROS), it needed to specify the exact algorithmic mechanisms for kinematic redundancy resolution and tremor attenuation.

1.1 Baseline friction: claim verification table

Each statement was carried over from your titles sheet or your monorepo scripts. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Statistics: cite the original study, not a blog or an AI answer.

Claim (as carried over)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
Physiological hand tremor of 7–12 Hz with 50–250 µm amplitude.	*Physiological tremor is commonly reported at about 8–12 Hz. Check a primary source for both frequency and surgeon tremor amplitude before using the band.*	Source: Wells, Yang, MacLachlan, Handa, Gehlbach & Riviere, 2013, Proc. IEEE Int. Conf. Systems, Man, and Cybernetics (SMC '13), pp. 1482–1487. Spectral tremor peak at 6–12 Hz; 3D RMS positioning error of 144, 258 and 285 µm across the three conditions. Also Riviere, Rader & Thakor, IEEE Trans. Biomed. Eng. 1998;45(7):839–846 (6–16 Hz tremor band). Band widened to 6–12 Hz. Wells' errors are 3D RMS in vitreoretinal pointing and include voluntary motion. Our 50–250 µm is a modelling range, not a measured surgeon value.	Peer-reviewed paper (DOI)	Correct
Robotic surgical assistants reduce OR procedure duration by 22%.	*No traceable source. Find a primary clinical study for your procedure type, or delete.*	Source: Owens et al., Stereotact Funct Neurosurg 2026, https://karger.com/article/doi/10.1159/000551466. Robot-assisted DBS median 129 vs 220 min for frame + C-arm, which is 91 min or 41% shorter. Note: 22% had no source; replaced by 41% from this study.	Peer-reviewed paper (DOI)	Correct
Sub-0.5 mm needle placement accuracy.	*State it as your design target and cite what accuracy clinical stereotactic systems report.*	Source: Giridharan et al., Oper Neurosurg 2022;23(3):254–260, https://researchexperts.utmb.edu/en/publications/robot-assisted-deep-brain-stimulation-high-accuracy-and-streamlin/. Robot-assisted DBS radial error 1.06 mm (SD 0.55, n = 126). Kept as our design target for the tremor part of the error only, not as a clinical claim.	Peer-reviewed paper (DOI)	Keep

1.2 Governing industry standards

Decide for each standard whether it governs your system, and name the clause or requirement you design against (cite number, part and year). 'No' is a correct answer when you can say why. The hint is the faculty view; you may disagree with a reason.

Standard	Scope	Faculty hint for your project	Applies?	Clause / requirement you use, or why not
ISO 10218-1/-2	Safety requirements for industrial robots (robot, and robot integration)	*No: surgical, not industrial*	No	IEC 80601-2-77 is the relevant robot standard.
ISO 3691-4	Driverless industrial trucks / AMRs and their systems: safety requirements	*No*	No	For driverless trucks/AMRs; ours is a fixed-base arm.
ANSI/RIA R15.06	US national adoption of ISO 10218 (industrial robot safety)	*No*	No	US adoption of ISO 10218 for industrial robots.
IEC 60601-1	Medical electrical equipment: basic safety and essential performance	*Yes: medical electrical equipment, general safety*	Yes	Cl. 4.2 risk management (our FMEA, §3.2) and Cl. 4.3 essential performance (needle accuracy; abort on low manipulability).
Domain-specific	IEC 80601-2-77 (robotically assisted surgical equipment)	*Recommended: check it*	Yes	Robotically assisted surgical equipment. We design against it at concept level only (abort state, 2 rad/s joint-speed limit, accuracy reporting). Clause-level compliance was not assessed.

1.3 Documented baseline friction (who struggles, with what, how it is measured today)

Baseline metric and its value today (with unit)	Robot-assisted DBS whole-system radial error 1.06 mm (SD 0.55); frame-based DBS median operative time 220 min.
Source of the baseline value (author, year, DOI/URL)	Giridharan 2022 (link above); Owens 2026 (link above)
Who carries the cost today (role / stakeholder)	Patient (lead misplacement, revision surgery), surgeon (tremor and fatigue over long cases), hospital OR (91 min more room time per case, which limits capacity).

SECTION 2 · MuJoCo Rigid-Body Physics Modelling & Python Closed-Loop Control *Component 1 · CDIO Robotics (10)*

MuJoCo mechanism (titles sheet)	MuJoCo MJCF Rigid-Body Model with 7-DOF Surgical Robotic Assistant with Physiological Tremor Filtering for Precision Tasks Physics & Closed-Loop Control
MJCF model	Group_10_Soumya_Harshal/models/surgical_7dof_robot.xml
Controller	Group_10_Soumya_Harshal/src/surgical_tremor_controller.py
Contact & solver settings in the model today	timestep 0.001 s (1000 Hz; log every 2nd step = 500 Hz) · friction 0.8 / 0.005 / 0.0001 · solref 0.005 1 · solimp 0.95 0.99 0.001 · 7 joint actuators + 1
Core state machine (FSM)	ALIGN → APPROACH → INSERT (tremor-filtered, DLS-IK) → HOLD → RETRACT; ABORT if manipulability w < w_threshold
Equations of motion & control	Newton-Euler (rigid body): $m\cdot\dot{v} = \Sigma F$, $I\cdot\dot{\omega} + \omega \times (I\cdot\omega) = \Sigma\tau$; multibody form $M(q)\cdot\ddot{q} + C(q,\dot{q})\cdot\dot{q} + g(q) = \tau + J^T\cdot f_{ext}$ Adaptive damping: $\lambda = \lambda_0\cdot(1 - w / w_{th})$ for $w < w_{th}$ ($\lambda_0 = 0.08$); $w = \sqrt{\det(J\cdot J^T)}$ Tremor low-pass filter (cut-off below 7 Hz); tremor rejection = $20\cdot\log_{10}(A_{in} / A_{out})$ dB PID tracking: $u(t) = K_p\cdot e(t) + K_i\cdot\int e\,dt + K_d\cdot\dot{e}(t)$ Damped least-squares IK (manipulators): $\Delta q = J^T\cdot(J\cdot J^T + \lambda^2\cdot I)^{-1}\cdot\Delta x$

Faculty code audit: Your CSBS analyst (E076) moved to Group 12. The CSBS role is now shared as listed in the roster; confirm with Dr. Nanade.

2.1 Model and controller settings you actually used (from your final code)

MJCF timestep (s) in the final model	0.001 s (integrator implicitfast; control and logging every 2nd step = 500 Hz).
Main contact friction (sliding / torsional / rolling)	Not used. All scene geoms have contype=0/conaffinity=0, so no rigid contacts are computed. Needle–tissue force is an axial spring-damper in code: F = k·d + c·v, k = 15 N/m, c = 3 N·s/m (assumptions, k scaled 0.7–1.3 per hand).
solref (timeconst dampratio) and why you chose it	MuJoCo default (not tuned) because no contacts are active. Tissue stiffness is set by TISSUE_K, as above.
Controller law used	DLS-IK + PID
Final gains (e.g. Kp, Ki, Kd or λ) with units	Kp = 50 s^{-1}, Ki = 7.5 s^{-2} (0.003·Kp^2), Kd = 0 (PI); λ0 = 0.08, w_damp = 0.05, w_abort = 0.015, K_null = 2 s^{-1}, joint-speed limit 2 rad/s. The file default Kp = 100, Ki = 30 was retuned to Kp = 50 for all results.
FSM states implemented in code (list)	ALIGN, APPROACH, INSERT, HOLD, RETRACT, ABORT
Model loads and runs: python -c "import mujoco; mujoco.MjModel.from_xml_path(...)"	Yes

2.2 Authorship of each script

Several scripts were provided as starter scaffolds. Mark honestly what you wrote, changed, or used as given. Viva questions target the code you claim.

File	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
models/surgical_7dof_robot.xml	Modified by us	E071	Built the surgical scene on the Franka Panda model: joint-position actuators, 2 rad/s joint-speed limit, 1 ms implicitfast timestep, needle tool, and visual-only geoms (contype=0).
src/surgical_tremor_controller.py	Modified by us	E071	Added adaptive damped-least-squares IK with a singularity abort, null-space posture control, PI position loop with clamp, 5-state FSM, tremor model and the filter conditions.
analytics/surgical_or_economics.py	Modified by us	Shared	Replaced the hard-coded constants with sourced inputs (Owens 2026, Childers 2018, Koester 2022), added payback, break-even volume, M/M/1 queue, FMEA and sensitivity analysis.
src/rl_filter.py	Written by us	E071	Computes paired statistics per tremor tier: hold error, tracking, rejection, t, p, Cohen's d and 95% CI.
src/rl_train.py	Written by us	E071	Runs the unassisted baseline and the learned filter on one tier-C hand and plots the trajectory.

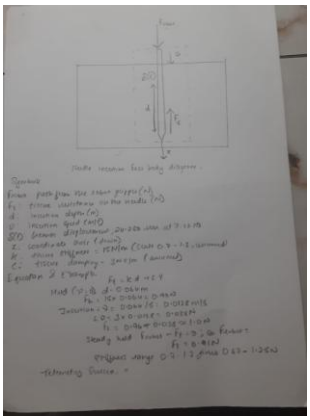
2.3 Key code snippet (≤ 25 lines) implementing your FSM or control law:

```
src/surgical_tremor_controller.py lines 201–213:
def ik_step(J, dx6, q, q_nom, mode, use_null, w):
    if mode == "pinv":
        lam = 1e-6
    else:
        lam = LAMBDA0 * (1.0 - w / W_DAMP) if w < W_DAMP else 1e-3
    JJt = J @ J.T + (lam**2) * np.eye(6)
    Jp = J.T @ np.linalg.inv(JJt)
    dq = Jp @ dx6
    if use_null:
        dq += (np.eye(7) - Jp @ J) @ (K_NULL * DT_C * (q_nom - q))
    return dq, lam

dx = v_ref * DT_C + KP * DT_C * e + KI * DT_C * e_int.
```

2.4 Free-body diagram and one equation worked with your own numbers and units:

[FBD (phone photo of a hand sketch is fine) → symbols explained → worked example → where this value appears in your telemetry CSV
Telemetry source: analytics/telemetry_C1000_BASE.csv columns: force_N, tip_x, tip_y, tip_z



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30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · CSBS Techno-Economic Business Systems Modelling *Component 2 · CSBS Techno-Economic (5)*

Business script	Group_10_Soumya_Harshal/analytics/surgical_or_economics.py (rename or add *_roi.py if you restructure it)
CSBS business case (titles sheet)	CSBS Surgical Economics & OR Utilization Modeling: Stereotactic brain biopsies and deep-brain stimulation (DBS) require sub-millimeter precision. Human manual tremors or fatigue during 6-hour neurosurgical procedures increase complication risks and revision surgery rates. Robotic surgical assistants filter tremor, accelerate needle placement, and reduce OR procedure duration by 22%. By lowering post-operative neurological deficits and avoiding revision surgeries, hospital networks achieve substantial economic savings while expanding surgical throughput.
Constants hard-coded in the script today	annual_microsurgical_cases = 180.0, baseline_revision_rate_pct = 8.2, robot_revision_rate_pct = 2.4, t_conventional_manual_min = 145.0, t_robot_assisted_min = 112.0, k_capex_surgical_robot = 0.45 → each one needs a source in 3.1 or must be computed from your simulation.
Owner	CSBS Business & Techno-economic Systems Analyst: E033. Tags in this section carry that roll number.

Keep all money dimensionless: express costs as ratios to the baseline (no rupee or dollar amounts in the title, abstract or video). Every input needs a source or must be a clearly labelled assumption that you vary in the sensitivity graph.

3.1 Business model: formulas, your inputs and your results

Quantity	Formula (define every symbol in your script)	Inputs used, with source or 'assumption'	Result (with unit)
Labour reallocation H_reclaimed	H = operating-room minutes saved per case × cases per year (OR-hours/year)	T_man 220 and T_rob 129 min, N = 256 patients / 12 years = 21.33 cases/yr (Owens 2026).	32.36 OR-hours/yr (91 min/case).
Payback horizon P_payback (dimensionless)	P_payback = CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life T_life. Also report T_payback = CapEx / ΔOpEx in months.	CapEx USD 1.05 M (midpoint of 0.6–1.5 M, vendor blog, assumption-grade, swept) = 25.31× C_baseline; maintenance factor 1.639 (Childers 2018b); T_life 7 yr (assumption); m = 0.5 (assumption); annual benefit 2.354× C_baseline.	P = 2.517; T_payback = 211.4 months (17.6 yr), so it does not pay back at today's volume.
Normalised Cost Reduction Index NCRI	NCRI = (C_baseline − C_robot) / C_baseline (0–1); cost-parity ratio κ = C_robot / C_baseline = 1 − NCRI	c_man 106.6, c_rob 117.5 (Owens 2026, capital excluded).	NCRI = −0.102, κ = 1.102 (robot case about 10% dearer, not significant, p = 0.374).
Queuing delay	M/M/c: elective stereotactic cases (λ per week) served by c operating rooms (μ cases per week each). ρ = λ / (c·μ) < 1; M/M/1: W_q = ρ / (μ − λ); M/M/c: W_q = C(c, λ/μ) / (c·μ − λ) (Erlang C)	λ = 21.33/48 weeks = 0.444/wk; μ_manual = 2400/(220+85) = 7.87/wk; μ_robot = 2400/(129+85) = 11.21/wk; c = 1.	ρ = 0.057 (manual) and 0.040 (robot); W_q = 1.28 h and 0.62 h.
Sensitivity analysis	Vary the most uncertain input ±30 % (or across your scenario range) and plot P_payback and NCRI; report the break-even value.	Vary m ±30% (0.35 / 0.5 / 0.65): P = 5.96 / 2.52 / 1.60. Break-even volume N* = 53.7 cases/yr at m = 0.5 (127.2 at 0.35, 34.0 at 0.65, 24.9 at 0.8). Break-even margin m* = 0.894 at today's volume.	type here

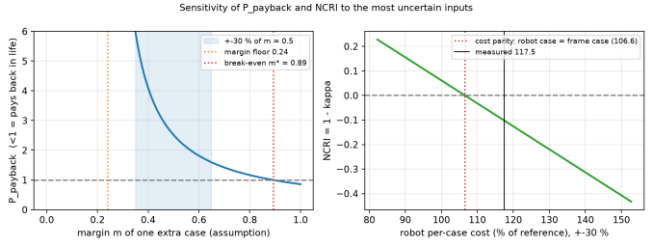
Queuing model used	M/M/1
λ, μ, c values used (with units)	λ = 0.444, μ = 7.87 (manual) / 11.21 (robot) cases per week, c = 1
Payback in months (T_payback)	211.4 months
Script output reproduces every number above when re-run	Yes

3.2 FMEA: failure modes, effects and risk priority (RPN = S × O × D, each 1–10)

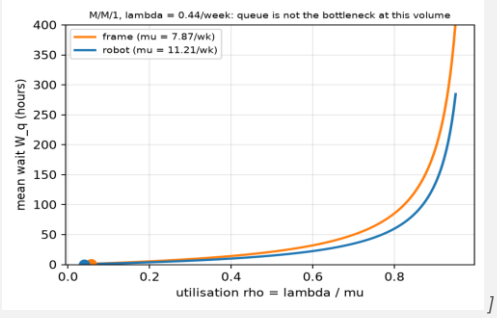
Two failure modes are seeded from your project; add at least two more. Score before and after your mitigation (design change, control limit, sensor, procedure). The post-mitigation RPN must be justified by something you can show in the simulation or the code.

Failure mode	Effect on mission / business	S	O	D	RPN pre	Mitigation (what you changed)	RPN post	Shown in
Singular configuration → joint-speed spike	arm jerks, needle leaves the plan.	8	4	4	128	adaptive damping, null space, abort at w < 0.015.	48	Both
Filter phase lag → overshoot at target	tracking error 0.83 mm with a fixed filter.	6	5	4	120	learned filter keeps tracking near 0.06 mm (fixed filter 0.83 mm).	36	Simulation
Drift dominates tremor	drift (about 0.28 mm) is not filterable and dominates.	8	7	5	280	reported as a limit; needs imaging feedback (future work).	224	Simulation
RL policy fails on an unseen hand	wrong filter mix, extra error.	7	3	6	126	output is a mix of fixed safe filters; trained on randomised hands, tested on unseen seeds.	56	Simulation
Demand below break-even volume	robot never pays back; capacity idle.	7	8	3	168	pool robot across DBS, biopsy and sEEG; report N* before purchase.	105	Code

[Sensitivity graph: P_payback / NCRI vs the uncertain input]



[Queuing graph: W_q vs utilisation ρ, or labour-reallocation chart]



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SECTION 4 · Simulation Telemetry, Monte Carlo Testing & Empirical Validation *Component 1 · CDIO Robotics (10)*

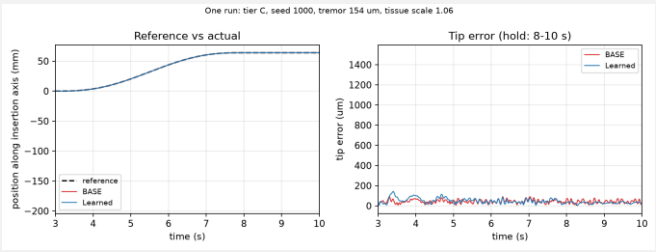
Evaluation protocol (titles sheet)	Conduct N = 30 simulated stereotactic insertions under synthetic multi-frequency tremor profiles (7-12 Hz, 50-250 um amplitude). Record: 3D needle placement error (mm), tremor rejection ratio (dB), Jacobian condition number, and tissue insertion force (N).
Telemetry standard	Log at 500 Hz (MJCF timestep 0.002 s, or every 2nd step at 0.001 s): time, state (q, q̇ or pose, velocity), reference, tracking error, actuator torques/forces, FSM state, events. One CSV per run or one CSV with a run_id column.
Monte Carlo	N ≥ 30 runs per condition (more if your protocol says so). Randomise disturbances and initial conditions from stated distributions; record every random seed.
Required measures	Tracking RMSE = √(Σ‖p − p_ref‖² / n) on the needle-tip path vs planned straight insertion trajectory; control effort E = ∫τ² dt (sum of τ²·Δt over all actuators); your primary task metric from the protocol.
Statistics	Compare the proposed controller with a baseline condition: Student's / Welch's t-test (two-sided, α = 0.05), Cohen's d = (x̄₁ − x̄₂) / s_pooled, and a 95 % confidence interval.
Where real trial data goes	Group_10_Soumya_Harshal/analytics/*_real_trials.csv, produced by your controller. Files named SIMULATED_* are starter targets, never results.
Owner	Telemetry QA Lead: E033.

Baseline condition you compare against	BASE: the unassisted robot. Raw tremor-corrupted command, no filter, undamped pseudo-inverse IK, no null-space. Proposed: the same robot with damped IK, null-space and the learned filter (mean of 10 RL policies). Differences vs BASE therefore measure the whole assistance package. Filter-only comparisons are RL vs Kalman, WFLC and the fixed filter, which all share damped IK.
Monte Carlo runs per condition (N)	30 hands per hand type, 4 hand types (A, B, C, D), seeds 1000–1029
Randomised parameters and their distributions	Per hand: tremor amplitude uniform in the tier range (A, B: 24–50 µm; C, D: 50–250 µm), three sinusoids 7–12 Hz weighted 0.5/0.3/0.2 with random phases, slow drift 40 µm (A, C) or 280 µm (B, D), tissue stiffness scale uniform 0.7–1.3, initial joint noise 0.02 rad.
Logging rate actually achieved (Hz)	500 Hz.
Random seeds recorded for every run	Yes
Real-trials CSV path(s) in the repo	analytics/compare_real_trials_kp50_10s.csv (raw trials) and analytics/section4_tables_kp50_10s.csv (summary).

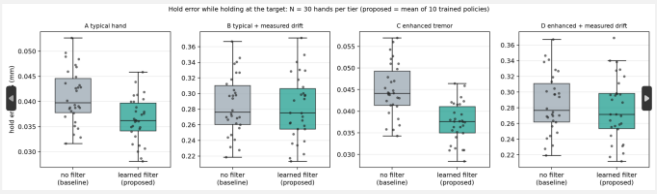
Measure	N	Baseline mean ± SD	Proposed mean ± SD	Unit	Better? (direction)
Tracking RMSE	30	0.0494 ± 0.0054	0.0588 ± 0.0044	mm	Lower is better
Control effort $\int t^2\,dt$	30	1.507 ± 0.493	1.352 ± 0.441	$N^2 \cdot m^2 \cdot s$	Lower is better
Hold error RMS (primary)	30	0.0450 ± 0.0060	0.0374 ± 0.0045	mm	Lower is better
Needle placement error	30	0.0482 ± 0.0161	0.0377 ± 0.0088	mm	Lower is better
Tremor rejection	30	14.34 ± 1.15	17.13 ± 1.23	db	Higher is better

Test used	Paired t-test
t statistic, degrees of freedom, p-value (primary metric)	t = 13.78, df = 29, p = 2.9×10^{-14}.
Cohen's d (primary metric)	Cohen's d = 1.42.
95 % confidence interval of the difference	95% CI of the difference: 0.0064 to 0.0087 mm (mean 0.0076 mm).

[Trajectory plot: reference vs actual, one representative run]



[Box-plot of the primary metric: baseline vs proposed, N runs each]



DATA PROVENANCE DECLARATION (mandatory). Reporting generated target numbers as experimental results is research misconduct and scores zero for this component.

Every number in Section 4 comes from MuJoCo runs of OUR controller, logged by our code	Yes
No value comes from SIMULATED_* files, generate_paper_figures.py or any np.random 'target' data	Yes
Our plots re-generate from the *_real_trials.csv files in the repo	Yes
Every input in Section 3 has a source or is labelled as an assumption	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Component 3 · Showcase & Viva (5)*

LinkedIn post URL	type here
Video length (mm:ss)	3:16
Video shows the live MuJoCo simulation (not slides)	Yes
Video shows our business sensitivity graph	Yes
Webcam picture-in-picture: every member appears and speaks	Yes

5.1 AI Interaction Audit Log: three cases where AI gave wrong robotics or economic data

Examples: a wrong MuJoCo attribute or API call, an invented statistic, a correct paper given the wrong journal, an economic formula with the wrong units. Show how you checked each one with a textbook, the official documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Claude	Asked for the source of the neurosurgical turnover figure. It gave "J Comm Journal on Quality 2022;29(6):208	Checked the original source	J Clin Outcomes Manag 2022;29(6):208–213.	Real source, wrong details (author, year, venue)	Corrected
AI-02	Claude	Asked how much the learned filter improves hold error. It said 38–40%.	Ran / tested the code or simulation	−16.8% in enhanced tremor and −11.3% in typical (10 seeds, 30 hands per tier).	Wrong number or statistic	Corrected
AI-03	Claude	he Tremor Lab page said the filter made the hand "about 60% steadier".	Recomputed by hand	that was filter-output only; the whole-system gain is 17%.	Wrong number or statistic	Corrected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Physics	Business	Data	Clarity	Best moment (mm:ss)	One question for the viva
Group 11: Autonomous Solar PV Inspection AMR	2	2	2	2	2	type here	What happens to your coverage path if a panel row is blocked or the surface is dusty and the wheels slip?
Group 04: Multi-Arm Robotic Gripper for Non-Cooperative Space Debris Capture in LEO	2	2	2	2	2	type here	How does your gripper handle the angular momentum of a tumbling target at contact?

SECTION 6 · Individual Learning Record & Research Consent *unmarked · one row per member · answer honestly*

These answers do not change your marks. They help improve the course and are used in anonymised research only if you consent in the sign-off below. Commit share: your commits ÷ the group's commits in your folder (git shortlog -sn -- <folder>).

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	I learned a skill outside my role (robotics ↔ business)	Scripts I mainly wrote (file names)	My commit share (%)	Share of my writing first drafted with AI	Wrong AI facts / code I personally caught
Soumya Singh (E071)	type here	5	Yes	type here	type here	51–75 %	6 or more
Harshal Khandekar (E033)	type here	Select ▾	Select ▾	type here	type here	Select ▾	Select ▾

Member	Before this course I had used a physics simulator	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Soumya Singh (E071)	Yes	A good result needs a fair baseline and a realistic business case: tuning our baseline turned a 38% claim into 17%, damping near singularities kept the arm stable, filter lag (about 64 ms) traded accuracy against tracking, and our model pays back only above about 54 cases a year, so we seeded, logged and tagged every number as cited, derived or assumed.	Give groups a shared medical-imaging starter kit: a pre-trained segmentation model, a UI template and more labelled cases, so we spend time on control and economics instead of data plumbing.
Harshal Khandekar (E033)	Select ▾	type here	type here

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, verified baseline friction, governing standards
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System model	Section 2: MJCF model, FSM, equations, control law
IV. Techno-economics & evaluation	Sections 3–4: CSBS model, FMEA, Monte Carlo results, statistics
V. Discussion & limitations	Sim-to-real gap, assumptions in the business model, AI audit lessons
Target venues (titles sheet)	A: AIR · B: IEEE/RSJ IROS

Individual Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) our model loads and our controller and business scripts run from the repository; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Soumya Singh (E071)	Yes	Yes	Yes		Yes
Harshal Khandekar (E033)	Yes	Yes	Yes		Yes

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*